3.3.2 Using the constant function with gradient descent

A constant mean function that is not zero is appropriate when we expect the objective function that we are modeling to take on some range of values that we know *a priori*. As we are modeling housing prices, using a constant mean function with a constant that is greater than zero makes sense, as we indeed expect the prices to be positive.

Of course, it's not possible to know what value the objective function takes on average in many cases. How, then, should we find an appropriate value for our mean function? The strategy we will use is to appeal to a specific quantity: how likely the training data set is, given the value for our mean function. Roughly speaking, this quantity measures how well our model explains its training data. We will see how to leverage this quantity to select the best mean function for our GP in this subsection.



If the likelihood of the training data given some value c_1 is higher than that given another value c_2, then we prefer using c_1 than using c_2. This quantifies our intuition above about using non-zero mean functions to model positive functions: a constant mean function whose value is positive explains observations from an exclusively positive function better than a function whose value is zero (or negative) does.

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